ML Model Deployment Strategies and Model Serving using FastAPI

1. Model Training and Evaluation

A Logistic Regression classification model was trained using the Titanic dataset. Preprocessing and classification were combined into a single pipeline. Model accuracy was evaluated using separate test records.

2. Model Saving

The complete trained pipeline was saved as model.pkl. The same saved pipeline was reused for all prediction methods without retraining.

3. Online Prediction

Online inference predicts one unseen passenger record per request. The FastAPI POST /predict endpoint returned the prediction and survival result.

4. Batch Prediction

Batch inference predicted 20 unseen passenger records together. The results were saved in batch_predictions.csv.

5. Streaming Prediction

Streaming inference simulated incoming passenger records by processing 10 records individually with a one-second delay between predictions.

6. FastAPI Model Serving

FastAPI exposed the trained model through REST endpoints. GET /health checked API status, and POST /predict returned passenger survival predictions.

7. Input Validation

Pydantic validated incoming passenger data. A valid request returned HTTP 200, while an invalid Pclass value of 5 returned HTTP 422.

8. Conclusion

The assignment demonstrated model training, evaluation, saving, online inference, batch inference, streaming simulation, API deployment, and request validation using a reusable ML pipeline.